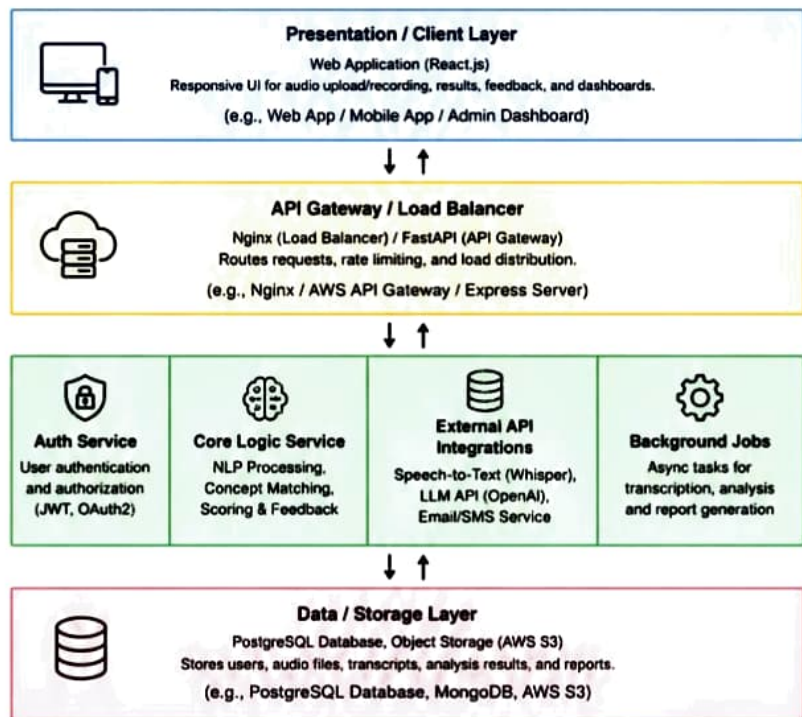


Solution Architecture

Date	5 July 2026
Team ID	–
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provides a high-level visual representation of your solution's architecture. Use the standard layered format to show how the user interacts with your system. It helps to clearly show what Data Layer, DB and Security technologies are components for your project directly into the boxes to create your architecture diagram



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Presentation Layer	Provides the user interface for students, teachers, and admins to interact with the system (upload audio, view scores, feedback, reports).	React.js, Tailwind CSS
API Gateway / Load Balancer	Routes incoming requests to appropriate services, handles load balancing, rate limiting, and security.	Nginx, FastAPI
Auth Service	Handles user registration, login, authentication, and authorization using secure tokens.	JWT, OAuth2.0, Python
Core Logic Service	Performs transcription, NLP analysis, concept matching, scoring, and generates feedback.	Python, FastAPI, spaCy, Transformers, Scikit-learn
External API Integrations	Integrates with third-party APIs for speech transcription, LLM, and notifications.	OpenAI API, Whisper API, SMTP / Twilio API
Background Jobs	Processes time-consuming tasks asynchronously (transcription, analysis, report generation).	Celery, Redis, RabbitMQ
Data / Storage Layer	Stores structured data, audio files, transcripts, analysis results, and generated reports.	PostgreSQL, AWS S3 (Object Storage)